

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS3

SUBJECT: Biology

TOPIC: Classification of Organisms

Introduction

- Classification is the grouping of organisms by shared features. It makes the study of life orderly and reveals evolutionary relationships. The modern hierarchy runs: kingdom → phylum/division → class → order → family → genus → species.

The five kingdoms

- **Monera** — bacteria and blue-green algae: prokaryotic (no true nucleus).
- **Protista** — protozoa (amoeba, paramecium) and algae: eukaryotic, single-celled or simple.
- **Fungi** — moulds, yeast, mushrooms: no chlorophyll, cell wall of chitin, saprophytic.
- **Plantae** — green plants: photosynthetic, cellulose walls.
- **Animalia** — heterotrophic, motile, no cell walls.
- Beyond the five: viruses are acellular — nucleic acid inside a protein coat; they are considered non-cellular particles, not placed in the kingdoms.

Binomial nomenclature

- Linnaeus: every organism has a genus and a species name in Latin — *Homo sapiens* (man), *Zea mays* (maize). The name is italicised (or underlined), the genus capitalised, the species in lower case.

Kingdoms in the exam

- Common examples: Monera — *Vibrio cholerae*; Protista — *Amoeba*, *Plasmodium*; Fungi — *Penicillium*, *Mucor*; Plantae — *Hibiscus*, *Pinus*; Animalia — *Chicken*, *Man*.

CLASS EVALUATION

1. List the five kingdoms with one feature of each.
2. What does binomial nomenclature mean?
3. In which kingdom do you place bacteria and why?
4. Why are viruses excluded from the five kingdoms?

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: EIGHT

CLASS: SS3

SUBJECT: Biology

TOPIC: Micro-organisms Around Us & Disease

Introduction

- Micro-organisms (microbes) are organisms too small to be seen with the naked eye: bacteria, protozoa, fungi, algae and viruses. They are found everywhere — in soil, water, air and in living bodies — and they may be useful or harmful.

Useful and harmful roles

- Useful: decomposition and nutrient cycling; nitrogen fixation; fermentation (bread, beer, yoghurt, garri); the production of antibiotics (penicillin from *Penicillium*); sewage treatment; vitamins in the gut. Harmful: they cause disease (pathogens), spoil food and toxify it (the aflatoxin of groundnuts), and cause decay.

Pathogenic micro-organisms

- **Bacteria** — cholera (*Vibrio cholerae*), typhoid (*Salmonella typhi*), tuberculosis (*Mycobacterium*), whooping cough.
- **Protozoa** — malaria (*Plasmodium*), amoebic dysentery (*Entamoeba histolytica*). **Viruses** — smallpox, polio, measles, HIV, COVID-19. **Fungi** — ringworm, athlete's foot, thrush. **Worms** (helminths) — roundworm, hookworm, tapeworm.

Transmission

- Air (droplets — coughs, sneezes), water (cholera, typhoid), food (contaminated — food poisoning), vectors (the female Anopheles mosquito for malaria; the tsetse fly for trypanosomiasis; houseflies carry pathogens), contact (skin, sexual contact — HIV, gonorrhoea), and contaminated instruments (needles, blades).

CLASS EVALUATION

1. State two useful and two harmful activities of micro-organisms.
2. Name the causative organisms of malaria and cholera.
3. How is typhoid spread?
4. Differentiate a pathogen from a vector with examples.

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: NINE

CLASS: SS3

SUBJECT: Biology

TOPIC: Micro-organisms & Public Health (Prevention)

Introduction

- Public health is the science of protecting and improving the health of communities — through sanitation, immunisation, safe water, health education and disease control. Micro-organisms are central to public health because they cause epidemics.

Methods of control

- Immunisation** — vaccines train the immune system (BCG, polio, measles, tetanus, yellow fever); **sanitation** — pit latrines, refuse collection, sewage treatment; **safe water** — boiling, chlorination, boreholes; **food hygiene** — washing, thorough cooking, covering; **vector control** — mosquito nets and indoor spraying (malaria), insecticide-treated bednets, draining stagnant water, fly proofing; **chemotherapy** — antibiotics for bacteria, antimalarials; and **health education** — hand washing, the danger of open defecation, and prompt treatment.

Diseases in Nigeria and their control

- Malaria (the highest killer), typhoid, cholera, tuberculosis, HIV/AIDS, measles and polio. The National Primary Health Care Development Agency runs immunisation and health programmes; the WHO and UNICEF support the campaigns.

The laboratory view

- Cultures on agar plates show colonies; the Gram stain distinguishes bacteria; microscopy and biochemical tests identify species; the spread-plate and pour-plate methods isolate pure cultures.

CLASS EVALUATION

1. Define public health.
2. State four ways of preventing the spread of cholera.
3. Name two diseases controlled by immunisation.
4. Why is mosquito control central to the fight against malaria?

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS3

SUBJECT: Biology

TOPIC: HIV/AIDS & Public Health

Introduction

- HIV (Human Immunodeficiency Virus) infects the helper T-cells of the immune system; AIDS is the terminal stage when the immune system collapses and opportunistic infections set in. It is a global public health issue and a manageable chronic disease with antiretroviral therapy (ART).

The biology of HIV

- HIV is a retrovirus: its genetic material is RNA, and the enzyme reverse transcriptase copies it into DNA which inserts into the host cell's genome. The virus is transmitted by body fluids — blood, semen, vaginal fluid and breast milk — not by casual contact, mosquitoes, or sharing food.

Progression and opportunistic infections

- The acute phase (flu-like symptoms), the asymptomatic phase (5–10 years, the CD4 count falls), and AIDS (CD4 below 200 cells/mm³, with opportunistic diseases — tuberculosis, pneumonia, candidiasis, Kaposi's sarcoma, wasting). Diagnosis: antibody tests (the rapid test, ELISA), with confirmatory Western blot or PCR.

Public health response in Nigeria

- The National Agency for the Control of AIDS (NACA) coordinates the response; prevention: ABC (Abstinence, Be faithful, Condoms), PMTCT (prevention of mother-to-child transmission), safe blood transfusion, harm reduction and testing; treatment: free ART, which suppresses the virus and prevents transmission (U=U — undetectable = untransmittable); and the reduction of stigma and discrimination.

CLASS EVALUATION

1. Explain why HIV is called a retrovirus.
2. State three routes of transmission and three non-routes.
3. What are opportunistic infections and why do they appear in AIDS?
4. What does U=U mean?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: ELEVEN

CLASS: SS3

SUBJECT: Biology

TOPIC: Economic Importance of Organisms

Introduction

- Organisms — plants, animals and micro-organisms — are of immense economic importance to the Nigerian economy: food, medicine, industry and the environment. This week surveys that importance and how it is managed.

Micro-organisms in industry

- Fermentation: garri, ogi, bread, beer, wine, yoghurt, cheese, soy sauce; antibiotics (penicillin, streptomycin); enzymes in detergents and food processing; insulin and vaccines by genetic engineering; biogas (methane) from waste; and sewage treatment.

Plants and animals of economic note

- Food crops (cassava, maize, rice, yam), cash crops (cocoa, oil palm, rubber, cashew, cotton), timber (mahogany, iroko), medicinal plants (neem, moringa, ginger); livestock (cattle, poultry, goats), fisheries and beekeeping; and the wild species that support tourism and the ecosystem (pollination, seed dispersal).

Harmful species and control

- Pests (stem borers, weevils, rodents), parasites (worms, ticks), pathogens, weeds and invasive species (water hyacinth blocking the Niger) — controlled by integrated pest management (IPM): resistant varieties, biological control, cultural practices and careful chemical use.

Biodiversity and conservation

- Ecosystem services (clean water, fertile soil, pollination) are worth more than the direct harvests; conservation — protected areas (national parks: Cross River, Yankari), the Endangered Species laws, sustainable fishing and forestry — preserves this capital for the future.

CLASS EVALUATION

1. State three industrial uses of micro-organisms.
2. Name four cash crops of Nigeria.

3. What is integrated pest management?
4. Why is biodiversity economically important?

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TWELVE

CLASS: SS3

SUBJECT: Biology

TOPIC: Micro-organisms in Biotechnology & Industry

Introduction

- Biotechnology is the use of living organisms or their products to make useful goods and services. Man has used micro-organisms for thousands of years — first unknowingly (bread, beer) and today by design (insulin, vaccines, GM crops).

Traditional biotechnology

- Fermentation — the conversion of sugar by micro-organisms: **yeast** makes bread (CO₂ leavens the dough), beer, wine, ogi and palm wine; **bacteria** (Lactobacillus) make yoghurt, cheese, soy sauce and the Nigerian 'ugba'; **moulds** ripen cheese and make soy sauce.

Modern biotechnology

- **Genetic engineering** — genes are transferred between organisms: human insulin and growth hormone are produced by bacteria; the hepatitis B vaccine in yeast; **transgenic crops** — pest-resistant maize (Bt), vitamin-A-enriched 'golden rice'. **Tissue culture** — plants (yam, cassava, banana) are multiplied from tiny pieces in sterile media, disease-free and fast. **Bioremediation** — micro-organisms are used to clean oil spills and sewage.

Applications and safety

- Medicine: antibiotics (penicillin from the mould *Penicillium*), vaccines, diagnostic kits; agriculture: biofertilisers (nitrogen-fixing *Rhizobium*), silage, the ripening of cheese; industry: citric acid, enzymes in detergents. Safety: the regulation of GMOs in Nigeria (the National Biosafety Management Agency), the ethics of gene editing, and the debate on genetically modified foods.

CLASS EVALUATION

1. Define biotechnology.
2. Name two micro-organisms and one product of each.
3. How is human insulin produced today?
4. What is tissue culture?
5. State two benefits and one risk of genetically modified crops.